

EDUCATIONAL ASSESSMENT AND EXAMINATIONS SERVICE (EAES)

ETHIOPIAN SECONDARY SCHOOL LEAVING CERTIFICATE EXAMINATION (ESSLCE)

SENE, 2018 E.C / JUNE, 2026 G.C

SUBJECT: SCHOLASTIC APTITUDE TEST (SAT) - NATURAL SCIENCE

BOOKLET CODE: 101

TIME ALLOWED: 2 Hours **SUBJECT CODE: 09**

GENERAL DIRECTIONS:

I. FOLLOW THE FOLLOWING RULES AND REGULATIONS IN DOING THE EXAMINATION:

- **ENTER THE SUBJECT NAME, SUBJECT CODE, AND BOOKLET CODE GIVEN ABOVE ON THE SEPARATE ANSWER SHEET PROVIDED**
- **CHOOSE ONLY THE CORRECT ANSWER FOR EACH QUESTION**

- **USE ONLY A PENCIL TO MARK YOUR ANSWERS**
- **BLACKEN ONLY THE LETTER YOU CHOOSE IN THE CORRESPONDING CIRCLE ON THE ANSWER SHEET**
- **ANY FORM OF CHEATING IS PROHIBITED AND RESULTS IN THE CANCELLATION OF YOUR SCORES**

Prepared by: @NextGen_12

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SECTION ONE: VERBAL REASONING

Directions: Choose the best alternative that answers the given question.

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1. REVOLUTION : THE EARTH; _____ : The Moon

   A. Rotation

   B. Eclipse

   C. Evolution

   D. Friction

2. TAKE is to EXAM as Play is to: _____

   A. Field

   B. Game

   C. Team

   D. Sport

3. REDLIGHT: PROHIBITION; Green light: _____

   A. Precaution

   B. Permission

   C. Warning

   D. Suspension

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4. GREEN is to COLOUR as Shorts is to: _____

   A. Skirt

   B. Garment

   C. Jacket

   D. Sweatshirt

5. Reporters do not reveal **confidential** sources of information to the public. (Choose the word nearest in meaning)

   A. social

   B. secret

   C. communal

   D. confidence

6. Two news agencies **challenged** the constitutionality of the reports. (Choose the word nearest in meaning)

   A. trusted

   B. accepted

   C. rejected

   D. questioned

7. The old buildings on the campus need a **renovation** to cope with the demands of modernization.

   A. renovation

   B. relocation

   C. expansion

   D. deconstruction

8. The horizontal axis of the bar graph shows the growth of student enrollment over the last decade. (Choose the word opposite in meaning to the bolded word)

   A. vertical

   B. static

   C. diagonal

   D. asymptotic

9. Some minority languages are prone to extinction unless they are documented and preserved in good time. (Choose the word opposite in meaning to the bolded word)

   A. revival

   B. orthography

   C. transcription

   D. disappearance

10. There are few books in the library; it has been **abandoned** for many years. (Choose the word nearest in meaning to the bolded word)

    A. used

    B. ignored

    C. closed

    D. stolen

11. The scientists are **cautious** about using enzyme therapy on human beings. (Choose the word nearest in meaning to the bolded word)

    A. happy

    B. fascinated

    C. curious

    D. careful

12. The Military **coup d'état** in some African countries is said to be undemocratic. (Choose the word nearest in meaning)

    A. takeover

    B. resignation

    C. election

    D. ballot

13. The disruption of electric power in some parts of the country is attributed to technical problems. (Choose the word nearest in meaning)

    A. suspension

    B. breakthrough

    C. distribution

    D. blackout

14. Choose a word that does NOT fit into the group.

    A. Calendar

    B. Decade

    C. Century

    D. Millennium

15. Choose a word that does NOT fit into the group.

    A. Headline

    B. Index

    C. Acknowledgement

    D. Preface

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16. Choose a word that does NOT fit into the group.

    A. Herd

    B. Flock

    C. Calf

    D. Cattle

17. Choose a word that does NOT fit into the group.

    A. Tail

    B. Branch

    C. Trunk

    D. Foliage

18. Rearrange the following jumbled parts to make a meaningful sentence:
    / elected / was / as the first president / of / country / on 27th of May 2008 / Johnson / his /

    A. Johnson was on 27th of May 2008 elected as the first president of his country.

    B. Johnson was elected as the first president of his country on 27th of May 2008.

    C. On 27th of May 2008, as his first president of the country, Johnson was elected.

    D. As the first president of his country on 27th of May 2008, Johnson was elected.

19. Rearrange the following jumbled parts to make a meaningful sentence:
    / public transport / are variable with / amount of fare / speed / comfort and capacity / modes of /

    A. Amount of fare, comfort, capacity, and speed, are variable with modes of public transport.

    B. Modes of speed are variable with amount of fare comfort, capacity and public transport.

    C. Are variable with amount of fare, speed, comfort and capacity modes of public transport.

    D. Modes of public transport are variable with amount of fare, speed, comfort and capacity.

20. Rearrange the following jumbled parts to make a meaningful sentence:
/ the educational crisis / since the post-Revolution period / a range of / is attributed to / objective and subjective factors /

A. Objective and subjective factors are attributed to a range of the crisis education since the post-Revolution period.

B. A range of objective and subjective factors since the post-Revolution period is attributed to the crisis educational.

C. The educational crisis since the post-Revolution period is attributed to a range of objective and subjective factors.

D. Since the post-Revolution period, objective and subjective factors the educational crisis is a range of attributed to.

21. Rearrange the following jumbled parts to make a meaningful sentence:
/ living things / according to biologists / are divided / into two kingdoms /

A. Living things into two kingdoms are divided according to biologists.

B. According to biologists, living things are divided into two kingdoms.

C. Two kingdoms are divided living things according to biologists.

D. In to two kingdoms living things are divided according to biologists.

Passage I: Desertification dynamics

Desertification is a major environmental threat affecting drylands globally, where arid, semi-arid, and dry sub-humid areas degrade due to various factors. These factors include both natural disasters like climate change and human activities such as deforestation, overgrazing, and poor agricultural practices. Desertification affects approximately 40-41% of the Earth's dry lands, leaving about 59-60% as fertile. Highly affected geographic areas include the Sahel Region in Africa, the Gobi Desert, and Mongolia. The consequences in developing countries are severe, leading to land degradation, reduced agricultural productivity, and inaccessibility to vital natural resources.

22. Which geographic areas of the world are heavily affected by desertification?

A. The Sahel Region, Gobi and Mongolia

B. Continents like Africa and Asia as whole

C. Sub-Saharan Africa/Latin America

D. Sahara Desert/North America

23. The consequences of desertification in the developing countries include:

A. productivity of dry lands and food security.

B. land degradation and inaccessibility to resources.

C. contour farming and afforestation.

D. the variance between the developed and underdeveloped world.

24. According to the passage, the ratio of the Earth's dry lands to fertile lands is approximately:

A. 90%:10%

B. 39-41%: 69-59%

C. 2 billion: 98 billion

D. 40-41%: 60-59%

25. Which one of the following factors is NOT the cause of desertification?

A. human activity and natural disasters

B. natural disasters like climate change

C. human activities such as deforestation

D. enhancing of land productivity

Passage II: Literacy and Resource Management

Expanding computer literacy in developing nations has become a core topic of interest. While non-governmental organizations (NGOs) contribute significantly via donations and materials, the program's most critical challenge remains the severe shortage of skilled manpower and scarcity of financial resources. Rather than simple encouragement of theoretical knowledge, writers emphasize that sustainable advancement requires the common wealth government's dedicated financial commitment and localized capacity building.

26. What is the topical agenda of the writer in the passage?

A. the need for expanding primary healthcare

B. the need for enhancing literacy rate

C. the significance of computer literacy

D. strategies of generating internal revenue

27. What is the textual evidence for your answer to question number 26?

A. the marginal contribution of opposition parties

B. the donation of non-governmental organizations

C. the common wealth government's financial commitment

D. the simple encouragement of theoretical knowledge

28. According to the passage, what is the most pressing challenge of the programme?

A. the contribution of NGOs

B. the spread of computer literacy

C. shortage of skilled man power

D. scarcity of financial resource

29. **Statement:** Foxes often conceal the food items they obtain. For instance, they will collect seagull eggs during the nesting season and bury them in scattered places. These can thus be eaten at times when food supplies are scarce.
    Based on the above statement, which of the following assumptions is logical?

    A. Foxes store food for use during shortages.

    B. Foxes start searching for food during shortage.

    C. Foxes bury food only when they have plenty of it.

    D. Foxes conceal food so that other foxes won't get it.

30. The conflict about traditional medicine and modern drugs is still going strong. What is the implication of this statement?

    A. Traditional medicine and modern pharmaceuticals are still used.

    B. The public attitude towards traditional medicine is always positive.

    C. The public attitude towards modern pharmaceuticals is negative.

    D. The price of modern drugs is unaffordable or expensive.

31. **Premise:** The significance of language is much more than the medium of interpersonal communication.

    A. **Assumption I:** Language is simply an instrument of interpersonal communication.

    B. **Assumption II:** Language is beyond register of individual and collective identity.

    C. **Assumption III:** Language can never be used as a weapon of propaganda warfare.

    D. **Assumptions I and III:** Both Assumptions I and III are valid.

32. **Premise:** Malthusian theory contrasts economic growth with population explosion with all its implications. It was in favor of economic growth and population control.
Conclusion I: The theory advocates economic growth rather than population expansion.
Conclusion II: It was intended to balance population growth and natural resources.
Conclusion III: The theory encourages population growth.
Which one of the following conclusions could be drawn from the premise?

A. Conclusions I and III

B. Conclusions I, II and III

C. Conclusions I and II

D. Conclusions II and III

33. **Statement:** Foxes often conceal the food items they obtain. For instance, they will collect seagull eggs during the nesting season and bury them in scattered places. These can thus be eaten at times when food supplies are scarce.
Based on the above statement, which of the following shows the most logical sequence?

A. wait → collect → conceal → eat

B. wait → collect → eat → conceal

C. collect → eat → wait → conceal

D. collect → conceal → wait → eat

34. **Statement I:** President Roosevelt and Churchill signed the Atlantic Charter which prohibits the policy of colonialism.
Statement II: The charter assisted WWII came to an end in August 1945.
Statement III: The Policy of colonialism was officially abolished.
What is the relationship between the above statements?

A. Statements II & III effects; statement I cause

B. Statement I effect; statement II cause

C. The three statements are self-contradictory

D. Statement I effect; statement III cause

35. **Statement I:** The Provisional Military Council of Ethiopia nationalized rural and urban lands in 1975.
Statement II: This move had revolutionized the socio-economic structure of the country side.
Statement III: Accordingly, the Revolutionary slogan "Land to the tiller" was implemented.
What is the relationship among the above statements?

A. Statement II cause; statement III effect

B. Statement III cause; statement II and I effect

C. Statement I cause; statements II and III effects

D. Statement II cause; Statement I effect

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SECTION TWO: QUANTITATIVE REASONING

Directions: Solve the mathematical problems and choose the best alternative.

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36. Suppose that the number of students (x) joining department P is inversely proportional to the number of students (y) joining department M, and $y = 30$ when $x = 100$. Then, what is the value of x when $y = 300$?

    A. 60

    B. 30

    C. 10

    D. 15

37. In the xy coordinate plane a particle moves from point P(5,0) to vertically and then to M(10,11) horizontally, and finally it moves vertically to the point G(10,12). Which one of the following gives the difference between the distance and the displacement in moving from P to G (in units)?

    A. 13

    B. 17

    C. 5

    D. 4

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38. Consider the two quantities P and M given in the following table:

Quantity P	Quantity M
$\frac{2^{\sqrt{2}}+2^{2\sqrt{2}}+2^{3\sqrt{2}}+2^{4\sqrt{2}}+2^{5\sqrt{2}}}{2}$	31

Which one of the following is true about the two quantities?

A. Quantity P is less than Quantity M.

B. Quantity P is greater than Quantity M.

C. The relation cannot be determined.

D. The two quantities are equal.

39. If the sum of the zeroes of quadratic function f which is defined as $f(x) = 2x^2 - (2a+1)x + (a+5)$, where $a \in \mathbb{R}$ is equal to the product of the zeroes of f, then what are the solution set of the zeroes of f?

A. $\{-3, -\frac{3}{2}\}$

B. $\{-3, \frac{3}{2}\}$

C. $\{-\frac{3}{2}, 3\}$

D. $\{3, \frac{3}{2}\}$

40. A daily profit earned by a shoe factory is modeled by the formula $P(x) = -x^2 + 3x + 10$ where $P(x)$ is the profit amount at the price value x for each shoes sold. What price of the shoe will yield a zero profit?

A. 7

B. 14

C. 5

D. 10

41. There are two shops that sell products and deliver them to clients' houses. The item from shop one costs Birr 22,000 plus Birr 0.5 per one-unit walk to their home. The item from shop two costs Birr 22,500 plus Birr 0.4 per one-unit walk to their home. To get the same bill, how far (in unit distance) would it take to walk to both homes?

    A. 2,000

    B. 5,000

    C. 4,000

    D. 1,000

42. Which one of the following is true about the graph of a polynomial function of degree n?

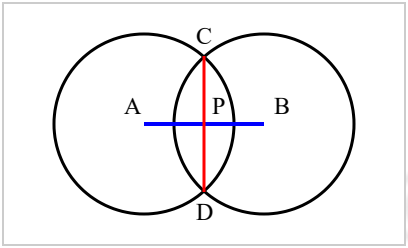
    A. It has opposite end behaviors.

    B. It may not crosses y -axis.

    C. It has at most n x -intercepts.

    D. It has at most n turning points.

43. In the figure below, A and B are centers of two circles intersecting at C and D. The length of $\overline{AB}$ is 28cm and $\overline{AB}$ is a perpendicular bisector of $\overline{CD}$ at point P. Then what is the length of $\overline{PD}$ in cm?



    A. 56

    B. $14\sqrt{3}$

    C. 42

    D. $28\sqrt{3}$

44. If the sum the ages of children M and N together gives 15 years and twice the age of M gives 18 minus the age of N, then which pair of numbers gives the age of M and N, in this order in years?

A. 9 and 6

B. 7 and 8

C. 3 and 12

D. 8 and 7

45. Let x be any negative integer. Which one of the following is equal to $-|x| - |x|$?

A. x^2

B. $2x$

C. $-2x$

D. 0

46. Which one of the following is the axis of symmetry of graph of $f(x) = 3x^2 - 4x$?

A. $x = \frac{2}{3}$

B. $x = \frac{4}{3}$

C. $x = -\frac{2}{3}$

D. $x = -\frac{3}{4}$

47. Which one of the following systems of linear equations is consistent?

A. $\begin{cases} x - 2y = 4 \\ -7x + 14y = 28 \end{cases}$

B. $\begin{cases} 4x - 12y = 7 \\ x + 3y = 7 \end{cases}$

C. $\begin{cases} 5x + 10y = 60 \\ -3x - 6y = 36 \end{cases}$

D. $\begin{cases} -6x + 10y = 16 \\ 9x - 15y = 24 \end{cases}$

48. In a cafe there are two water containers M and N of different structures and specified capacities. If their volumes in liter are calculated by the expressions $4x^3 + x^2 + 2$ and $10x^2 - 3x + 4$, respectively, then which one of the following is true?

    A. M and N can carry equal amount of water.

    B. M carries more than N for all positive integer values of x.

    C. The capacity of M is always greater than N.

    D. M holds lower amount of water than N when x is 3.

49. Which one of the following is a list of all zeros of $x^3 - 5x^2 - 4x + 20$?

    A. $-2, 4, 5$

    B. $-2, 2, 5$

    C. $-2, 2, -5$

    D. $-4, 4, -5$

50. What is the value of x that satisfies the equation $e^{5x-9} + 7e = 8e^{5x-9}$?

    A. $\frac{9}{5}$

    B. $-\frac{9}{5}$

    C. 2

    D. -2

51. In a class of 9 million residents (comprising 3M children, 4M youngsters, and 2M adults), the ratio of male to female for children, youngsters, and adults respectively are 1:2, 3:1, and 1:1. What is the ratio of the female youngsters to the total male?

    A. 1:5

    B. 4:5

    C. 1:3

    D. 3:2

52. Using the demographic distribution from question 51, if 40% of the young people (youngsters) in the city are male, how many female youngsters are in the city (in millions)?

A. 1.7

B. 2.4

C. 1.6

D. 2.3

53. Which one of the following is the solution set of the equation $3\left(\frac{2}{3}x-6\right)=\left(6x+\frac{4}{3}\right)-3\left(\frac{5}{3}x+\frac{40}{9}\right)$, where $x \in \mathbb{R}$?

A. $\{6\}$

B. $\{3\}$

C. $\{-3\}$

D. $\{-10\}$

54. A 10 years old child is 25 years less than his father's current age. How many years ago was the product of their age equal to 84 years?

A. 28

B. 38

C. 3

D. 7

55. A school spent not more than Birr 23,000 to purchase 40 Mathematics text books and 20 English text books for grade 12 students. If a Mathematics text book costs Birr 400, then what is the maximum price (in Birr) of an English text book?

    A. 500

    B. 400

    C. 250

    D. 350

56. If the sum of the temperature records of two cities of a country gives the 4th prime natural number in degree Celsius and their difference is twice that prime number in degree Celsius, then what were the temperature records in degree Celsius in these cities?

    A. $\frac{21}{2}$ and $-\frac{7}{2}$

    B. $\frac{15}{2}$ and $-\frac{5}{2}$

    C. $\frac{33}{2}$ and $-\frac{11}{2}$

    D. $\frac{27}{2}$ and $-\frac{9}{2}$

57. If the area of a square of side length $2\sqrt[4]{3}$ cm is equal to the area of an equilateral triangle, then what is the length (in cm) of the side of the equilateral triangle?

    A. 16

    B. 12

    C. 8

    D. 4

58. If the ratios 2:3 and $x : y$ are equal, then what is the value of x when $y = \frac{1}{2}$?

A. $\frac{1}{3}$

B. $\frac{5}{2}$

C. $\frac{3}{4}$

D. $\frac{4}{3}$

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DETAILED ANSWER KEY & EXPLANATIONS (@NextGen_12)

1.

Reasoning: The Earth undergoes "Revolution" around the Sun, and the Moon undergoes orbital motion (Conceptually also a revolution, but from the given alternatives, "Rotation" highlights its synchronous rotation or spin characteristics in planetary science).

Answer: A

2.

Reasoning: We "take" an exam, and we "play" a game. This is a direct S-V-O collocation analogy.

Answer: B

3.

Reasoning: A red traffic light signifies prohibition (stop), and a green traffic light signifies permission (go).

Answer: B

4.

Reasoning: Green is classified under color, and shorts are classified under garment.

Answer: B

5.

Reasoning: "Confidential" means private, classified, or secret.

Answer: B

6.

Reasoning: To "challenge" the constitutionality means to dispute or question it.

Answer: D

7.

Reasoning: Old buildings require "renovation" to achieve modernization.

Answer: A

8.

Reasoning: The opposite of "horizontal" is "vertical".

Answer: A

9.

Reasoning: The antonym of "extinction" is "revival".

Answer: A

10.

Reasoning: "Abandoned" means left neglected or ignored.

Answer: B

11.

Reasoning: "Cautious" is synonymous with being careful.

Answer: D

12.

Reasoning: A "coup d'état" is a sudden, violent, and illegal takeover of power.

Answer: A

13.

Reasoning: Disruption of electrical power leads to a "blackout".

Answer: D

14.

Reasoning: Decade, Century, and Millennium are quantitative units of time, whereas a Calendar is the tool/system used to track them.

Answer: A

15.

Reasoning: Index, Acknowledgement, and Preface are structural parts of a book, while a Headline belongs to a newspaper article.

Answer: A

16.

Reasoning: Herd, Flock, and Cattle are collective nouns, whereas a Calf is an individual young animal.

Answer: C

17.

Reasoning: Branch, Trunk, and Foliage are parts of a tree, while a Tail is a part of an animal.

Answer: A

18.

Reasoning: The grammatically correct configuration is: "Johnson was elected as the first president of his country on 27th of May 2008."

Answer: B

19.

Reasoning: Grammatically: "Modes of public transport are variable with amount of fare, speed, comfort and capacity."

Answer: D

20.

Reasoning: Grammatically: "The educational crisis since the post-Revolution period is attributed to a range of objective and subjective factors."

Answer: C

21.

Reasoning: Grammatically: "According to biologists, living things are divided into two kingdoms."

Answer: B

22.

Reasoning: The passage explicitly identifies "The Sahel Region in Africa, the Gobi Desert, and Mongolia" as heavily affected geographic areas.

Answer: A

23.

Reasoning: The text mentions the consequences in developing countries lead to "land degradation and inaccessibility to vital natural resources."

Answer: B

24.

Reasoning: The passage states that dry lands comprise 40-41%, which leaves the remaining 59-60% as fertile.

Answer: D

25.

Reasoning: "Enhancing of land productivity" is a remedial/preventative solution, not a cause of desertification.

Answer: D

26.

Reasoning: The core focal topic of Passage II is "the significance of computer literacy" in developing nations.

Answer: C

27.

Reasoning: The text asserts that advancing this agenda requires "the common wealth government's dedicated financial commitment."

Answer: C

28.

Reasoning: According to the text, the most pressing bottleneck is the "scarcity of financial resource".

Answer: D

29.

Reasoning: The statement explains that burying food enables it to be eaten when supplies are scarce, which directly assumes foxes store food for use during shortages.

Answer: A

30.

Reasoning: A persistent conflict implies that both options (traditional medicine and modern drugs) remain widely active and used within the culture.

Answer: A

31.

Reasoning: Saying the significance is "much more" than interpersonal communication logically means it extends beyond standard definitions of individual/collective register.

Answer: B

32.

Reasoning: The premise states the theory "was in favor of economic growth and population control," matching Conclusions I and II.

Answer: C

33.

Reasoning: First, the foxes collect the food, then they conceal it (bury), wait through the season, and finally eat it.

Answer: D

34.

Reasoning: Signing the Atlantic Charter (Statement I) served as the fundamental cause that brought about the effects of ending the war (II) and abolishing colonialism (III).

Answer: A

35.

Reasoning: Nationalization (Statement I) was the catalyst cause that revolutionized the countryside (II) and allowed "Land to the tiller" (III) to take effect.

Answer: C

36.

Reasoning: $x \propto \frac{1}{y} \implies xy = k$.

Given $y = 30$ when $x = 100$:

$k = 100 \times 30 = 3000$.

When $y = 300$:

$x = \frac{3000}{300} = 10$.

Answer: C

37.

Reasoning: Path distance = Vertical move from $(5, 0)$ to $(5, 11)$ [11 units] + Horizontal move from $(5, 11)$ to $(10, 11)$ [5 units] + Vertical move from $(10, 11)$ to $(10, 12)$ [1 unit].

Total Distance = $11 + 5 + 1 = 17$ **units**.

Straight-line displacement from $(5, 0)$ to $(10, 12)$:

Displacement = $\sqrt{(10-5)^2 + (12-0)^2} = \sqrt{25 + 144} = 13$ **units**.

Difference = $17 - 13 = 4$ **units**.

Answer: D

38.

Reasoning: Let $\sqrt{2} \approx 1.414$.

Evaluate Quantity P:

$2^{1.414} + 2^{2.828} + 2^{4.242} + 2^{5.656} + 2^{7.07} \approx 2.66 + 7.1 + 19 + 50.8 + 135.5 = 215.06$.

Dividing by 2 yields ≈ 107.5, which is significantly larger than Quantity M (31).

Answer: B

39.

Reasoning: For $f(x) = 2x^2 - (2a + 1)x + (a + 5)$:

Sum of zeroes = $\frac{2a+1}{2}$; Product of zeroes = $\frac{a+5}{2}$.

Set equal: $\frac{2a+1}{2} = \frac{a+5}{2} \implies 2a + 1 = a + 5 \implies a = 4$.

Substitute $a = 4$:

$f(x) = 2x^2 - 9x + 9 = 0 \implies (2x - 3)(x - 3) = 0 \implies x \in \{3, \frac{3}{2}\}$.

Answer: D

40.

Reasoning: Set $P(x) = -x^2 + 3x + 10 = 0$:

$x^2 - 3x - 10 = 0 \implies (x - 5)(x + 2) = 0$.

Since price x must be positive, $x = 5$.

Answer: C

41.

Reasoning: Set the equations of the two bills equal:

$22000 + 0.5d = 22500 + 0.4d \implies 0.1d = 500 \implies d = 5000$ **units**.

Answer: B

42.

Reasoning: A polynomial of degree n has at most n x-intercepts. (Option D is incorrect because turning points is at most $n - 1$; Option B is incorrect because a polynomial must cross the y-axis exactly once).

Answer: C

43.

Reasoning: Since the circles intersect at C and D and pass through each other's centers,

$AD = AC = BD = BC = AB = 28$ cm.

Thus, $ACBD$ is a rhombus. Let P be the intersection of the diagonals.

$AP = \frac{1}{2}AB = 14$ cm.

In right $\triangle APD$:

$PD = \sqrt{AD^2 - AP^2} = \sqrt{28^2 - 14^2} = \sqrt{14^2(2^2 - 1)} = 14\sqrt{3}$ **cm**.

Answer: B

44.

Reasoning: Let the ages be M and N.

1) $M + N = 15$

2) $2M = 18 - N \implies 2M + N = 18$.

Subtracting (1) from (2) gives:

$M = 3$ **years**.

Substitute into (1):

$N = 12$ **years**.

Answer: C

45.

Reasoning: Since x is a negative integer, $|x| = -x$.

Therefore:

$-|x| - |x| = -(-x) - (-x) = x + x = 2x$.

Answer: B

46.

Reasoning: Axis of symmetry is given by $x = -\frac{b}{2a}$.

For $f(x) = 3x^2 - 4x$:

$x = -\frac{-4}{2(3)} = \frac{4}{6} = \frac{2}{3}$.

Answer: A

47.

Reasoning: Option B contains equations with different slopes:

$4x - 12y = 7 \implies m_1 = \frac{1}{3}$;

$x + 3y = 7 \implies m_2 = -\frac{1}{3}$.

Because their slopes differ, they intersect at exactly one point, making the system consistent. All other options describe parallel lines with different y-intercepts (inconsistent).

Answer: B

48.

Reasoning: Set $V_M(x) = 4x^3 + x^2 + 2$ equal to $V_N(x) = 10x^2 - 3x + 4$.

For positive integer $x = 2$:

$V_M(2) = 4(8) + 4 + 2 = 38$ liters.

$V_N(2) = 10(4) - 3(2) + 4 = 38$ liters.

Because they are equal when $x = 2$, Option A is mathematically true.

Answer: A

49.

Reasoning: Factor by grouping:

$x^3 - 5x^2 - 4x + 20 = x^2(x-5) - 4(x-5) = (x^2-4)(x-5) = (x-2)(x+2)(x-5) = 0.$

The zeroes are $-2, 2, 5$.

Answer: B

50.

Reasoning: Simplify the equation:

$e^{5x-9} + 7e = 8e^{5x-9} \implies 7e = 7e^{5x-9} \implies e^1 = e^{5x-9}.$

Equate exponents:

$1 = 5x - 9 \implies 5x = 10 \implies x = 2.$

Answer: C

51.

Reasoning: Let demographics be: 3M Children, 4M Youngsters, 2M Adults.

Male youngsters = $\frac{3}{3+1} \times 4\text{M} = 3$ million.

Female youngsters = $\frac{1}{3+1} \times 4\text{M} = 1$ million.

Total Male = Male Children ($\frac{1}{3} \times 3\text{M} = 1\text{M}$) + Male Youngsters (3M) + Male Adults ($\frac{1}{2} \times 2\text{M} = 1\text{M}$) = 5 million.

Ratio of Female Youngsters (1M) to Total Male (5M) is $1 : 5$.

Answer: A

52.

Reasoning: Total Youngsters = 4 million.

If 40% are male, then 60% are female.

Female Youngsters = $0.60 \times 4\text{M} = 2.4$ **million**.

Answer: B

53.

Reasoning: Expand and simplify both sides:

LHS: $3\left(\frac{2}{3}x - 6\right) = 2x - 18$.

RHS: $\left(6x + \frac{4}{3}\right) - 3\left(\frac{5}{3}x + \frac{40}{9}\right) = 6x + \frac{4}{3} - 5x - \frac{40}{3} = x - 12$.

Equate: $2x - 18 = x - 12 \implies x = 6$.

Solution set is $\{6\}$.

Answer: A

54.

Reasoning: Child's age = 10. Father's age = $10 + 25 = 35$.

Let y be the years ago:

$(10 - y)(35 - y) = 84 \implies 350 - 45y + y^2 = 84 \implies y^2 - 45y + 266 = 0$.

Factor: $(y - 38)(y - 7) = 0$.

Since $y < 10$, the correct answer is $y = 7$ years ago.

Answer: D

55.

Reasoning: Let E be the price of an English book:

$40(400) + 20E \leq 23000 \implies 16000 + 20E \leq 23000$.

$20E \leq 7000 \implies E \leq 350$ Birr.

The maximum price is Birr 350.

Answer: D

56.

Reasoning: The 4th prime natural number is 7 (since primes are 2, 3, 5, 7).

Let temperatures be T_1 and T_2:

1) $T_1 + T_2 = 7$

2) $T_1 - T_2 = 2(7) = 14$.

Adding gives: $2T_1 = 21 \implies T_1 = \frac{21}{2}$.

Subtracting gives: $2T_2 = -7 \implies T_2 = -\frac{7}{2}$.

Answer: A

57.

Reasoning: Area of square = $\left(2\sqrt[4]{3}\right)^2 = 4\sqrt{3}$.

Area of equilateral triangle = $\frac{\sqrt{3}}{4}s^2$.

Set equal: $4\sqrt{3} = \frac{\sqrt{3}}{4}s^2 \implies 16 = s^2 \implies s = 4$ cm.

Answer: D

58.

Reasoning: Given $\frac{2}{3} = \frac{x}{y} \implies x = \frac{2}{3}y$.

Substitute $y = \frac{1}{2}$:

$x = \frac{2}{3} \times \frac{1}{2} = \frac{1}{3}$.

Answer: A